Elliot Blackstone, Ph.D.

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SUMMARY

Computer vision and machine learning engineer with a Ph.D. in mathematics and hands-on experience developing, training, optimizing, and deploying deep learning models in PyTorch. Built custom convolutional and transformer-based object detection systems and optimized inference through quantization, ONNX deployment, and custom C++/CUDA kernels. Published mathematical research in top-tier peer-reviewed journals, bringing a strong foundation in mathematical modeling, optimization, and analytical problem solving to computer vision.

SKILLS & CERTIFICATIONS

- Programming & Tools: Python (NumPy, pandas, scikit-learn, matplotlib), C++, LaTeX, Jupyter, Git, Mathematica, Linux
- ML & Computer Vision: PyTorch, torchvision, CUDA, CNNs, transformers, ONNX, Quantization, OpenCV
- Tabular ML: XGBoost, LightGBM, CatBoost, Random Forest, linear/logistic regression, Optuna, SHAP
- Deployment & MLOps: Docker, FastAPI, Google Cloud Platform (Cloud Run), ONNX Runtime
- Certifications/Awards: Erdős Institute Data Science Boot Camp, UofM Honored Instructor, UCF best dissertation award

SELECTED PROJECTS

Automotive Object Detection [Online Demo] [GitHub] **2026**

- Built a from-scratch Single Shot Detector in PyTorch for automotive object detection on a 30k-image dataset with 195k labeled bounding boxes; improved data augmentation, sampling, and training pipeline to reach **0.54 mAP@0.50** on the test set.
- Re-architected the project into a modular training and inference pipeline and eliminated bottlenecks through profiling, mixed precision, batched target matching, batched NMS, and DataLoader tuning, reducing end-to-end training time from **50h to 25h**.
- Developed gen-nms-package, a standalone **PyTorch C++/CUDA extension** for GIoU, DIoU, and CIoU non-maximum suppression on CPU and GPU, enabling compiled postprocessing beyond the standard torchvision NMS operators.
- Built optimization and deployment stack around the model, including ONNX export, PyTorch/ONNX Runtime parity testing, static INT8 post-training quantization, benchmarking, and both browser-based and local real-time webcam/video inference.
- Designed and trained a from-scratch transformer-based object detector in PyTorch with 11.3M parameters, achieving 0.73 mAP@0.50 on the test set while outperforming the 24.5M-parameter SSD baseline across all five object classes.

Predicting Calorie Expenditure - *The Erdős Institute* [LINK] [GitHub] **2025**

- Built a supervised learning pipeline to predict workout calorie expenditure from a 750k-row tabular dataset, with log-transform and engineered features, training and tuning linear, GAM, XGBoost, LightGBM, CatBoost, ensemble, and AutoGluon models using Optuna and SHAP, and achieving **4th place out of 4,318 teams** in a Kaggle competition.

WORK EXPERIENCE

University of Michigan: Ann Arbor, MI **2021 - 2025**

Postdoctoral Assistant Professor and James Van Loo Postdoctoral Fellow

- Implemented numerical simulations and data analysis pipelines in Mathematica to study PDE dynamics and visualize emergent patterns. This work led to the publication of four papers and was presented at international conferences.
- Developed course material and taught over 300 students topics such as probability theory, calculus, linear algebra, polynomial regression, and mathematical modeling. Won a teaching award and was named UofM Honored Instructor.

KTH Royal Institute of Technology: Stockholm, Sweden **2019 - 2021**

Postdoctoral Researcher

- Conducted full-time research in random matrix theory. Performed numerical analysis to verify asymptotic convergence rates, ensuring theoretical models aligned with computational outputs. This work led to the publication of four papers.

University of Central Florida: Orlando, FL **2012 - 2019**

Research & Teaching Assistant

- Co-authored two papers as part of PhD research. Created rigorous methods for ill-posed inverse problems in computed tomography, using spectral theory and asymptotic analysis to characterize reconstruction stability. Won best dissertation award.
- Developed course material and taught over 600 students between 12 calculus courses, two differential equations courses, and two linear algebra courses. Topics taught include linear regression, singular value decomposition, and multivariable calculus.

EDUCATION

University of Central Florida, *Ph.D. Mathematics* **2019**
University of Central Florida, *M.S. Mathematics* **2014**
Penn State Erie, *B.S. Mathematics (Statistics Minor)* **2011**